Discrete Mathematical structures [December 2020]

(MAT 203)

PART-A

1. Without using truth tables, show that

$p \to (q \to r) \equiv p \to (\sim q \vee r) \equiv (p \wedge q) \to r$

Ans:-

$$p \to (q \to r) \equiv p \to (\sim q \vee r)$$
$$\equiv (\sim p) \vee (\sim q \vee r)$$
$$\equiv \sim (p \wedge q) \vee r$$
$$\equiv (p \wedge q) \to r$$

2. Define the terms: Converse, Inverse and contrapositive

Ans: The converse of $p \to q$ is the implication $q \to p$

The inverse of $p \to q$ is the implication $\sim p \to \sim q$

The contrapositive of $p \to q$ is the implication $\sim q \to \sim p$.

3. What is pigeonhole principle? Given a group of 100 people, at minimum, how many people were born in the same month?

Ans:- If m pigeons occupy n pigeonholes, $m > n$, then atleast one pigeonhole will contain more than one pigeon.

$\left\lceil \frac{100}{12} \right\rceil = 9$ people were born in the same month.

4. In how many ways can the letters of the word 'MATHEMATICS' be arranged such that vowels must always come together?

Ans: The word has 11 letters.

If all vowels are considered as one total 8 letters.

$\therefore$ no. of arrangements $= \frac{8!}{2!\,2!} \times \frac{4!}{2!} = 120960$

5. If $A = \{1, 2, 3, 4\}$, give an example of a relation on A which is reflexive and transitive, but not symmetric.

Ans: $R = \{(1,1), (2,2), (3,3), (4,4), (1,2)\}$

6. Define a complete lattice. Give an example.

Ans: A lattice is complete lattice if each of its non empty subset have lub and glb.

eg: $(D_{12}, |)$ is a complete lattice. Here lub is 12 and glb is 1.

7. Define a recurrence relation. Give an example.

Ans: The recurrence relation for a sequence $\{a_n\}$ is an equation that express the general term of the sequence $\{a_n\}$ in terms of one or more of the previous terms in the sequence namely $a_0, a_1, a_2 \ldots a_{n-1} \ldots$

eg: Let $\{a_n\} = \{3, 6, 9, 12, \ldots\}$

This is a geometric sequence with common ratio 3

So $a_{n+1} = 3a_n,\ n \geq 0,\ a_0 = 3$ is the recurrence relation

8. Determine the coefficient of x^{15} in

$f(x) = (x^2 + x^3 + x^4 + \cdots)^4$

Ans. $(x^2 + x^3 + x^4 + \cdots)^4 = x^8 (1 + x + x^2 + \cdots)^4$

$= x^8 (1-x)^{-4}$

The cofft of x^{15} in $f(x)$ is same as the cofft of x^7 in $(1-x)^{-4}$

$(1-x)^{-4} = 1 + 4x + \frac{4.5}{2!} x^2 + \frac{4.5.6}{3!} x^3 + \frac{4.5.6.7}{4!} x^4 + \cdots$

$\therefore$ cofft of $x^{15} = \frac{4.5.6.7.8.9.10}{7!}$

9. Define a semi-group. Give an example.

Ans: If S is a non-empty set and $*$ is a closed binary operation on S, and $*$ is associative on S, then the algebraic system $(S, *)$ is a semi-group.

eg: The set Z^+ of all +ve integers is a semigroup under the binary operation addition.

10. Show that the set of idempotent elements of any commutative monoid forms a submonoid.

Ans: Let $(M, *, e)$ be a commutative monoid.

Let A be the set of idempotent elements of M.

We're to S.T i) A is closed ii) $A \subset M$ iii) $e \in A$.

We've $a * a = a$, $b * b = b$ $\forall a, b \in A$.

Consider $(a*b) * (a*b) = (a*b) * (b*a)$ ($\because M$ is abelian)

$= a * (b*b) * a$ ($*$ is associative)

$= a * b * a$ ($\because b$ is idempotent)

$= a * a * b$ (M abelian)

$= a * b$ ($\because a$ is idempotent

$\therefore a * b \in A$.

Thus A is closed w.r.t $*$ and $A \subseteq M$.

We know that $e * e = e$, $e \in A$.

Thus $(A, *, e)$ is a submonoid of $(M, *, e)$

Part B

Module 1

11 a) Check whether the propositions $p \wedge (\sim q \vee r)$ and $p \vee (q \wedge \sim r)$ are logically equivalent or not.

Soln:

p	q	r	$\sim q$	$\sim r$	$q \wedge \sim r$	$\sim q \vee r$	$p \wedge (\sim q \vee r)$	$p \vee (q \wedge \sim r)$
1	1	1	0	0	0	1	1	1
1	1	0	0	1	1	0	0	1
1	0	1	1	0	0	1	1	1
1	0	0	1	1	0	1	1	1
0	1	1	0	0	0	1	0	0
0	1	0	0	1	1	0	0	1
0	0	1	1	0	0	1	0	0
0	0	0	1	1	0	1	0	0

Truth table of $p \wedge (\sim q \vee r)$ and $p \vee (q \wedge \sim r)$ are not the same. So the propositions are not logically equivalent.

11 b) Check the validity of the argument

$p \rightarrow q$

$q \rightarrow r \wedge s$

$\sim r \vee (\sim t \vee u)$

$p \wedge t$

$\therefore u$

Soln:

Steps	Reasons
1) $p \to q$	Premise
2) $q \to r \wedge s$	Premise
3) $p \to r \wedge s$	steps 1 and 2 and Law of syllogism
4) $p \wedge t$	Premise
5) p	step 4 and the Rule of conjunctive simplification
6) $r \wedge s$	steps 3 and 5 and the Rule of Detachment
7) r	step 6 and the Rule of Conjunctive simplification
8) $\neg r \vee (\neg t \vee u)$	Premise
9) $\neg (r \wedge t) \vee u$	step 8, the Associative law and De Morgan's Laws.
10) t	step 4 and the Rule of conjunctive simplification
11) $r \wedge t$	step 7 and 10 and the Rule of conjunctio
12) $\therefore u$	Rule of Disjunctive syllogism.

12 a) show that $((p \to q) \wedge (q \to r)) \to (p \to r)$ is a tautology

soln:

p	q	r	$p \to q$	$q \to r$	① $(p \to q) \wedge (q \to r)$	② $p \to r$	① $\to$ ②
1	1	1	1	1	1	1	1
1	1	0	1	0	0	0	1
1	0	1	0	1	0	1	1
1	0	0	0	1	0	0	1
0	1	1	1	1	1	1	1
0	1	0	1	0	0	1	1
0	0	1	1	1	1	1	1
0	0	0	1	1	1	1	1

$\therefore ((p \to q) \wedge (q \to r)) \to p \to r$ is a Tautology.

b) Let p, q, r be the statements given as

p: Arjun studies , q: He plays cricket , r: He passes Data structures

Let P_1, P_2, P_3 denote the premises

P_1: If Arjun studies then he will pass data structures

P_2: If he doesn't play cricket then he will study

P_3: He failed data structures.

Determine whether the argument $(P_1 \wedge P_2 \wedge P_3) \to q$ is valid.

Soln:

$P_1 : p \rightarrow r$

$P_2 : \neg q \rightarrow p$

$P_3 : \neg r$

				P_1	P_2	P_3	$P_1 \wedge P_2 \wedge P_3$	$P_1 \wedge P_2 \wedge P_3 \rightarrow q$
p	q	r	$\neg q$	$p \rightarrow r$	$\sim q \rightarrow p$	$\sim r$		
1	1	1	0	1	1	0	0	1
1	1	0	0	0	1	1	0	1
1	0	1	1	1	1	0	0	1
1	0	0	1	0	1	1	0	1
0	1	1	0	1	1	0	0	1
0	1	0	0	1	1	1	1	1
0	0	1	1	1	0	0	0	1
0	0	0	1	1	0	1	0	1

$(P_1 \wedge P_2 \wedge P_3) \rightarrow q$ is a Tautology.

$\therefore$ Argument is valid.

PART B (Module II)

13 (a) State Binomial theorem. Find the coefficient of xyz^2 in $(2x-y-z)^4$.

Binomial theorem states that:

A: • If x and y are variables and n is a positive integer, then

$$(x+y)^n = {}^nC_0\, x^0 y^n + {}^nC_1\, x^1 y^{n-1} + {}^nC_2\, x^2 y^{n-2} + \dots + {}^nC_n\, x^n y^0$$

$$= \sum_{k=0}^{n} {}^nC_k\, x^k y^{n-k}.$$

• Coefficient of xyz^2 in $(2x-y-z)^4$

$a = 2x$, $b = -y$, $c = -z$

$\therefore$ the coefficient of abc^2 in $(a+b+c)^4$ is $\binom{4}{1\ 1\ 2}$.

But $\binom{4}{1\ 1\ 2}(abc^2) = \dfrac{4!}{1!\,1!\,2!}\,(2x)(-y)(-z)^2$

$= \dfrac{4!}{2!}\; 2 \times (-1)(-1)^2\, xyz^2$

$= 4 \times 3 \times 2 \times -1\; xyz^2$

$= -24\, xyz^2$

$\therefore$ coefficient of xyz^2 is -24

(b) Determine the number of positive integers n such that $1 \le n \le 100$ and n is not divisible by 2, 3 or 5.

A: $S = \{1, 2, 3, \dots 100\}$

$N = 100$

C_1 : n is divisible by 2

C_2 : n is divisible by 3

C_3 : n is divisible by 5

$$N(C_1) = \left\lfloor \frac{100}{2} \right\rfloor = 50$$

$$N(C_2) = \left\lfloor \frac{100}{3} \right\rfloor = \lfloor 33.3 \rfloor = 33$$

$$N(C_3) = \left\lfloor \frac{100}{5} \right\rfloor = 20$$

$$N(C_1C_2) = \left\lfloor \frac{100}{\text{lcm}(2,3)} \right\rfloor = \left\lfloor \frac{100}{6} \right\rfloor = \lfloor 16.6 \rfloor = 16$$

$$N(C_1C_3) = \left\lfloor \frac{100}{\text{lcm}(2,5)} \right\rfloor = \left\lfloor \frac{100}{10} \right\rfloor = 10$$

$$N(C_2C_3) = \left\lfloor \frac{100}{\text{lcm}(3,5)} \right\rfloor = \left\lfloor \frac{100}{15} \right\rfloor = \lfloor 6.6 \rfloor = 6$$

$$N(C_1C_2C_3) = \left\lfloor \frac{100}{\text{lcm}(2,3,5)} \right\rfloor = \left\lfloor \frac{100}{30} \right\rfloor = 3$$

$$\therefore N(\bar{C_1}\bar{C_2}\bar{C_3}) = N - [N(C_1) + N(C_2) + N(C_3)] + [N(C_1C_2) + N(C_1C_3) + N(C_2C_3)] - N(C_1C_2C_3)$$

$$= 100 - [50+33+20] + [16+10+6] - 3$$

$$= 100 - 103 + 32 - 3 = 26$$

There are 26 numbers which are not divisible by 2, 3 or 5.

14 (a) Prove that if 7 distinct numbers are selected from $\{1, 2, 3, \ldots, 11\}$ then sum of two among them is 12.

A: Let m = pigeons = 7 distinct numbers selected from $\{1, 2, 3, \ldots 11\}$.

n = pigeonholes = $\{1, 11\}$, $\{2, 10\}$, $\{3, 9\}$, $\{4, 8\}$, $\{5, 7\}$, $\{6\}$

= 6 sets whose sum of elements is 12.

By pigeonhole principle if m pigeons fly to n pigeonholes and $m > n$ then atleast one pigeoholes must contain ~~atleast~~ 2 or more pigeons occupying in it.

(b) An urn contains 15 balls, 8 of which are red and 7 are black. In how many ways can 5 balls be chosen so that

(i) all the five are red

(ii) all the five are black

(iii) 2 are red and 3 are black

(iv) 3 are red and 2 are black.

A: No. of balls = 15

No. of red balls = 8

No. of black balls = 7

(i) Choosing 5 red balls from 8 red balls = ${}^{8}C_{5}$ = 56

(ii) Choosing 5 black balls from 7 black balls = ${}^{7}C_{5}$ = 21

(iii) Choosing 2 red and 3 black balls = ${}^{8}C_{2} \times {}^{7}C_{3}$ = 980

(iv) Choosing 3 red and 2 black balls = ${}^{8}C_{3} \times {}^{7}C_{2}$ = 1176

Module 3

15 a) If $f, g,$ and h are functions on integers, $f(n) = n^2$, $g(n) = n+1$, $h(n) = n-1$; then find 1) $f \circ g \circ h$, 2) $g \circ f \circ h$ 3) $h \circ f \circ g$

soln:

$$f \circ g \circ h = f(g(h(n))) = f(g(n-1)) = f(n-1+1)$$
$$= f(n) = n^2$$
$$g \circ f \circ h = g(f(h(n))) = g(f(n-1)) = g((n-1)^2)$$
$$= (n-1)^2 + 1$$
$$h \circ f \circ g = h(f(g(n))) = h(f(n+1)) = h((n+1)^2)$$
$$= (n+1)^2 - 1 = n^2 + 2n$$

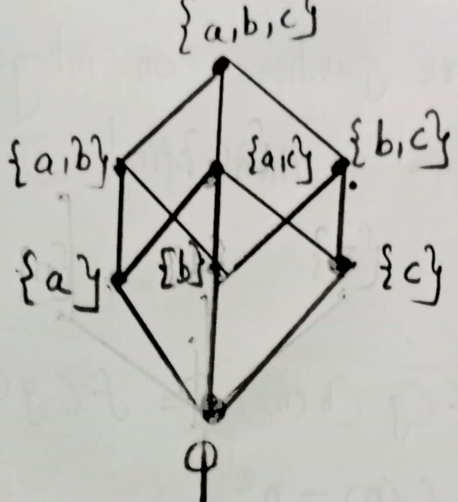
b) If $A = \{a, b, c\}$ and $P(A)$ be its power set. The relation $\leq$ be the subset relation defined on the powerset. Draw the Hasse diagram of $(P(A), \leq)$

soln:

$$P(A) = \{\phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{a,c\}, \{b,c\}, \{a,b,c\}\}$$

$$R = \{(\phi, \phi), (\phi, \{1\}), \dots (\phi, \{1,2,3\})$$
$$(\{1\}, \{1\}), (\{1\}, \{1,2\}), \dots (\{1\}, \{1,2,3\})$$
$$(\{2\}, \{2\}), (\{2\}, \{1,2\}) \dots (\{2\}, \{1,2,3\})$$
$$\vdots$$
$$(\{1,2,3\}, \{1,2,3\})\}.$$

Hasse Diagram



16 a) Let R be a relation on Z by xRy if $4/(x-y)$. Then find all equivalence classes.

soln: $R = \{ (x,y),\ x,y \in Z \text{ and } 4/(x-y)\}$

$= \{ (x,y),\ x,y \in Z \text{ and } x-y = 4k,\ k \in Z\}$

$[0] = \{ \ldots -8, -4, 0, 4, 8, \ldots \}$

$= \{4k,\ k \in Z\}$

$[1] = \{ \ldots -7, -3, 1, 5, \ldots \} = \{4k+1,\ k \in Z\}$

$[2] = \{ \ldots -6, -2, 2, 6 \ldots \} = \{4k+2,\ k \in Z\}$

$[3] = \{ \ldots -5, -1, 3, 7, \ldots \} = \{4k+3,\ k \in Z\}$.

b) Find the complement of each element in D_{42}

soln: $D_{42} = \{1, 2, 3, 6, 7, 14, 21, 42\}$

Greatest element $I = 42$

Least element $O = 1$

$$a_n = a_n^{(h)} + a_n^{(p)} \quad \text{——} \ 1$$

$$= c_1 2^n - 1$$

$$a_0 = 0 \Rightarrow c_1 2^0 - 1 = 0$$

$$\Rightarrow c_1 = 1 \quad \text{——} \ 1$$

$$\therefore a_n = 2^n - 1, \ n \geq 0 \quad \text{——} \ 1$$

17b) Solve the recurrence relation

$$a_{n+2} = a_{n+1} + a_n, \ n \geq 0, \ a_0 = 0, \ a_1 = 1$$

Ans: characteristic eqn is $r^2 - r - 1 = 0$

$$\Rightarrow r = \frac{1 \pm \sqrt{5}}{2} \quad \text{——} \ 2$$

$\therefore$ the general soln, $a_n = c_1\left(\frac{1+\sqrt{5}}{2}\right)^n + c_2\left(\frac{1-\sqrt{5}}{2}\right)^n$ —— 2

$$a_0 = 0 \Rightarrow c_1 + c_2 = 0 \Rightarrow c_2 = -c_1$$

$$a_1 = 1 \Rightarrow c_1\left(\frac{1+\sqrt{5}}{2}\right) - c_1\left(\frac{1-\sqrt{5}}{2}\right) = 1$$

$$\frac{c_1}{2}\left[1+\sqrt{5} - 1 + \sqrt{5}\right] = 1$$

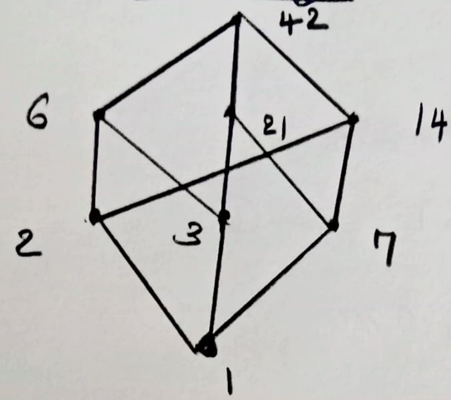
$$\frac{2\sqrt{5}}{2} c_1 = 1 \Rightarrow c_1 = \frac{1}{\sqrt{5}} \quad \text{——} \ 1$$

$$\therefore c_2 = \frac{-1}{\sqrt{5}} \quad \text{——} \ 1$$

$$\therefore a_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n, \ n \geq 0$$

—— 2

Hasse Diagram



lub $\{1, 42\} = I = 42$

glb $\{1, 42\} = 0 = 1$

$\therefore 1' = 42$ and $42' = 1$

lub $\{2, 21\} = 42$

glb $\{2, 21\} = 1$

$\therefore 21' = 2$ and $2' = 21$

lub $\{3, 14\} = 42$

glb $\{3, 14\} = 1$

$\therefore 3' = 14$ and $14' = 3$

lub $\{7, 6\} = 42$

glb $\{7, 6\} = 1$

$\therefore 7' = 6$ and $6' = 7$.

Module - IV

17 a) Solve the recurrence relation

$a_{n+1} = 2a_n + 1, \ n \geq 0, \ a_0 = 0.$

Ans: characteristic eqn is $r - 2 = 0$

$\Rightarrow r = 2$

$\therefore a_n^{(h)} = c_1 2^n$ —— (2)

$a_n^{(P)} = B.1^n$

$a_{n+1} - 2a_n = 1$

$B - 2B = 1 \Rightarrow B = -1$

$\therefore a_n^{(P)} = -1.$ —— (1)

18a) Solve the recurrence relation

$a_{n+2} - 4a_{n+1} + 3a_n = -200, \; n \geq 0, \; a_0 = 3000, \; a_1 = 3300$

Ans: char. eqn is $r^2 - 4r + 3 = 0$

$$r = 3, 1 \quad — 1$$

$$\therefore a_n^{(h)} = c_1 3^n + c_2 1^n \quad — 1$$

R.H.S $\cancel{a_n^{(p)}} = -200 . 1^n$

$$\therefore a_n^{(p)} = B . n . 1^n$$

Sub. $a_n^{(p)}$ in the given eqn,

$$B(n+2) - 4B(n+1) + 3Bn = -200$$

$$Bn + 2B - 4nB + 4B + 3Bn = -200$$

$$\Rightarrow -2B = 100 \Rightarrow B = 100$$

$$\therefore a_n^{(p)} = 100n. \quad — 1$$

$$a_n = a_n^{(h)} + a_n^{(p)}$$

$$= C_1 3^n + C_2 1^n + 100n \quad — 1$$

$a_0 = 3000 \Rightarrow C_1 + C_2 = 3000 \quad — (1)$

$a_1 = 3300 \Rightarrow 3C_1 + C_2 + 100 = 3300$

$$3C_1 + C_2 = 3200 \quad — (2)$$

Solving eqns (1) & (2),

$$2C_1 = 200 \Rightarrow C_1 = 100$$

$$C_2 = 2900 \quad — 1$$

$$\therefore a_n = 100 \, 3^n + 2900 + 100n, \; n \geq 0 \quad — 1$$

18b) Solve the recurrence relation

$$a_n = 2a_{n-1} - 4a_{n-2},\ n \geq 3,\ a_1 = 2,\ a_2 = 0$$

Ans: Char. eqn is $r^2 - 2r + 4 = 0$

$$\Rightarrow r = 1 \pm i\sqrt{3} \quad \text{— ①}$$

$$\therefore a_n = c_1(1+i\sqrt{3})^n + c_2(1-i\sqrt{3})^n \quad \text{— ①}$$

$$1 + i\sqrt{3} = r(\cos\theta + i\sin\theta)$$

$$r = \sqrt{x^2+y^2} = 2,\quad \theta = \tan^{-1}(y/x)$$
$$\tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \quad \text{— ①}$$

$$1 + i\sqrt{3} = 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) \quad \text{— ①}$$

$$1 - i\sqrt{3} = 2\left(\cos\left(\frac{\pi}{3}\right) - i\sin\frac{\pi}{3}\right) \quad \text{— ①}$$

$$a_n = c_1\left[2\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right]^n + c_2\left[2\left(\cos\frac{\pi}{3} - i\sin\frac{\pi}{3}\right)\right]^n$$
$$= c_1\left[2^n\left(\cos\frac{n\pi}{3} + i\sin\frac{n\pi}{3}\right)\right] + c_2\left[2^n\left(\cos\frac{n\pi}{3} - i\sin\frac{n\pi}{3}\right)\right]$$
$$= 2^n\left[k_1\cos\frac{n\pi}{3} + k_2\sin\frac{n\pi}{3}\right], \quad \text{— ①}$$

where

$$k_1 = c_1 + c_2$$
$$k_2 = i(c_1 - c_2)$$

$$a_1 = 2 \Rightarrow 2\left(k_1\cos\frac{\pi}{3} + k_2\sin\frac{\pi}{3}\right) = 2 \Rightarrow 2\left(\frac{k_1}{2} + \frac{k_2\sqrt{3}}{2}\right) = 2$$

$$a_2 = 0 \Rightarrow 4\left(k_1\cos\frac{2\pi}{3} + k_2\sin\frac{2\pi}{3}\right) = 0 \Rightarrow 4\left(-\frac{k_1}{2} + k_2\frac{\sqrt{3}}{2}\right) = 0$$

On simplifying, $k_1 = 1$, $k_2 = \frac{1}{\sqrt{3}}$ — ①

$$\therefore a_n = 2^n\left(\cos\frac{n\pi}{3} + \frac{1}{\sqrt{3}}\sin\frac{n\pi}{3}\right) \quad \text{— ①}$$

$$\frac{k_1}{2} + k_2\frac{\sqrt{3}}{2} = 1$$
$$-\frac{k_1}{2} + k_2\frac{\sqrt{3}}{2} = 0$$
$$2k_2\frac{\sqrt{3}}{2} = 1$$
$$k_2 = \frac{1}{\sqrt{3}}$$
$$\Rightarrow k_1 = 1$$

19 a) If $f: (R^+, \cdot) \to (R, +)$ as $f(x) = \ln x$ where R^+ is the set of positive real no's. Show that f is a monoid isomorphism from R^+ onto R.

Soln:

For all $a, b \in R^+$

$f(a \cdot b) = \ln(ab) = \ln a + \ln b = f(a) + f(b)$

$f(1) = \ln(1) = 0$

$\therefore f$ is a monoid homomorphism.

f is one-to-one since $f(a) = f(b)$

$\Rightarrow \ln a = \ln b$

$\Rightarrow e^{\ln a} = e^{\ln b} \Rightarrow a = b$

f is onto since for each $x \in R$ there exist $e^x \in R^+$ such that $\ln(e^x) = x$

$\therefore f$ is a monoid isomorphism.

b) Show that every subgroup of a cyclic group is cyclic.

Proof: Let $G = \langle a \rangle$

If H is a subgroup of G, then each element of H has the form a^k for some $k \in Z$

For $H \neq \{e\}$, Let t be the smallest positive integer such that $a^t \in H$.

To prove H is a cyclic group, we have to prove that $H = \langle a^t \rangle$

Since $a^t \in H$, by the closure property for the subgroup H, $\langle a^t \rangle \subseteq H$ —— ①

Now we want to prove that $H \subseteq \langle a^t \rangle$

Let $b \in H$, $b = a^s$ for some $s \in \mathbb{Z}$

$S = qt + r$ [By division algorithm], $q, r \in \mathbb{Z}$

$0 \leq r < t$

$a^s = a^{qt+r} = a^{qt} \cdot a^r$

$a^r = a^s \cdot a^{-qt} = a^s \cdot (a^t)^{-q} \in H$

[Since $a^t \in H$ and H is a subgroup $\therefore (a^t)^{-q} \in H$
$a^s \in H$ and $(a^t)^{-q} \in H$ $\therefore$ by closure property
$a^s \cdot (a^t)^{-q} \in H$]

But if $a^r \in H$, with $r > 0$, then it contradicts the minimality of t $\therefore r = 0$ $\therefore b = a^{qt} = (a^t)^q \in \langle a^t \rangle$

$\therefore H \subseteq \langle a^t \rangle$ —— ②

From eqn ① and ② $H = \langle a^t \rangle$.

20 a) state and prove Lagrange's Theorem.

Lagrange's Theorem

If G is a finite group of order n with H a subgroup of order m, then m divides n.

Proof: If $H = G$, the result follows

otherwise if $m < n$, there exist an element

$a \in G - H$.

since $a \notin H \Rightarrow aH \neq H \Rightarrow aH \cap H = \phi$

If $G = aH \cup H$ then $|G| = |aH| + |H| = 2|H|$,

the result follows.

If not there exist an element $b \in G - (H \cup aH)$

with ~~b∩∈~~ $bH \cap H = \phi = bH \cap aH$ and $|bH| = |H|$

If $G = bH \cup aH \cup H \Rightarrow |G| = |bH| + |aH| + |H| = \cancel{3H}$
$= |H| + |H| + |H| = 3|H|$

If not there exist an element $c \in G - (bH \cup aH \cup H)$

The group G is finite so this process terminates and

$G = a_1H \cup a_2H \cup \ldots \cup a_kH$

$\therefore |G| = k|H|$

20 b) If $A = \{1, 2, 3\}$ List all permutations on A and prove that it is a group.

soln:

$S_3 = \{ f_1, f_2, f_3, f_4, f_5, f_6 \}$

$f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$; $f_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$; $f_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$

$f_4 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$; $f_5 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$; $f_6 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$

$\cdot$	f_1	f_2	f_3	f_4	f_5	f_6
f_1	f_1	f_2	f_3	f_4	f_5	f_6
f_2	f_2	f_3	f_1	f_6	f_4	f_5
f_3	f_3	f_1	f_2	f_5	f_6	f_4
f_4	f_4	f_5	f_6	f_1	f_2	f_3
f_5	f_5	f_6	f_4	f_3	f_1	f_2
f_6	f_6	f_4	f_5	f_2	f_3	f_1

From the table closure property, Associative property are true.

$f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$ is the identity element

$f_1^{-1} = f_1$; $f_2^{-1} = f_3$; $f_3^{-1} = f_2$, $f_4^{-1} = f_4$, $f_5^{-1} = f_5$

$f_6^{-1} = f_6$

$\therefore (S_3, \cdot)$ is a group.